Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Substitution: units used = power × time (1) i.e. power is {2400 **or** 2.4} **and** time is {2.5 **or** $\left(\frac{2.5}{60}\right)$ **or** (2.5 × 6) **or** $\left(\frac{15}{60}\right)$} Correct answer i.e. 0.6 [kWh] (1) Award 1 mark for answers only of 0.1, 6, 36, 100, 600, 6000 and 36 000	1	1		2	2	
		(ii)		Cost = 0.6 **ecf** × 7 × 34 (1) Cost = 142.8 [p] (1) accept 143 [p] Award 1 mark for answer of 20.4 [p] **or** 4.2	1	1		2	2	
		(iii)		Any calculation that shows that power × time is the same for both kettles (1) e.g. for kettle A: units used = $\frac{2}{60} \times 3 \times 6 = 0.6$ So cost is the same so Sophia is correct (1) Conclusion mark can only be awarded if the calculation is present and is consistent with (a)(i) **Alternative:** 3 × 0.033 × 6 = 0.594 [kWh] (1) The cost is similar so agree with Sophia **or** the cost is less so disagree with both of them (1) Conclusion mark can only be awarded if the calculation is present and is consistent with (a)(i)			2	2	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		**Any 2 × (1) from:** – the cables can be made thinner – each part of the cable carries less current – sockets can be placed anywhere on the ring – [each socket has 230 V applied] they can be operated separately	2			2		
		(ii)		Faster [acting] (1) accept more sensitive Can be {reset / reused} **or** doesn't need replacing (1)	2			2		
		(iii)		Total power = 7750 W [so he is not correct]			1	1		
				Question 5 total	**6**	**2**	**3**	**11**	**6**	**0**